OBJ00000000

ECTIVES

- To review basic operational amplifier applications and explore some of their limitations.
- To learn good techniques for circuit building, testing, and debugging.

EQUIPMENT

- Breadboard and various wires/jumper
- Various resistors (1/4 watt; 100 Ω, 1kΩ, 10kΩ, 100kΩ)
- Diligent Analog Discovery (DAD) Board
- LM741CN Op Amp
- CA3130EZ Op Amp

INTRODUCTION

Operational amplifiers (op-amps) are fundamental building blocks in analog electronic circuits and are often analyzed under the assumption of ideal behavior in this class. In the ideal model, op-amps are assumed to have infinite open-loop gain, infinite input impedance, zero output impedance, infinite bandwidth, and perfect rejection of common-mode signals. Real op-amps, however, deviate from these assumptions in ways that can significantly affect circuit performance.

This laboratory experiment focuses on examining the non-ideal characteristics of practical op-amps and evaluating their impact on real-world circuit behavior. Key non-idealities include finite open-loop gain, finite, limited common-mode rejection ratio (CMRR), input offset voltages and bias currents, slew rate limitations, and output voltage and current constraints.

Understanding op-amp non-idealities is essential for accurate circuit design, analysis, and troubleshooting in practical applications. Through this lab, you will gain insight into how these imperfections influence amplifier gain, bandwidth, linearity, and signal integrity, reinforcing the importance of considering real-world device characteristics when designing analog circuits.

Pre-Lab

PART 1: Non-Inverting and Inverting Amplifiers (10 Points)

An inverting operational amplifier will produce an output that is 180 degrees out of phase from the input voltage. While non-inverting operational amplifiers will be in phase with no phase difference.

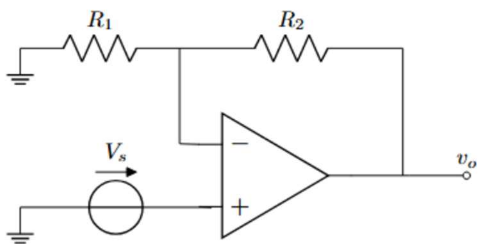


Figure 1 – Non-Inverting Configuration

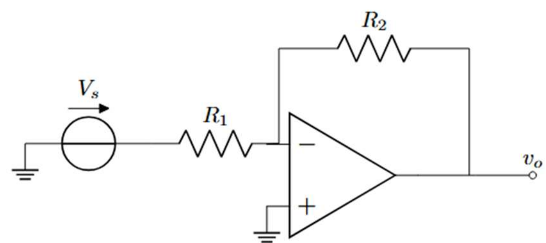


Figure 2 – Inverting Configuration

Figures 1 and 2 show the non-inverting and inverting voltage amplifiers to be constructed. **Initially the LM741CN op amp will be used**, but later it will be replaced with a CA3130EZ op amp (they have similar pinouts) so avoid running wires over the op amp chip.

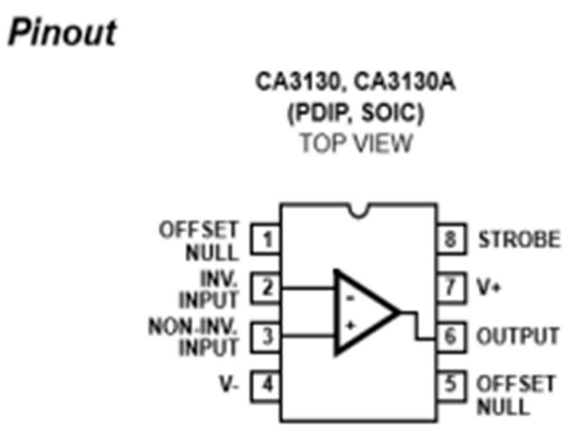


Figure 3 – CA3130 or LM741 Pin-Out

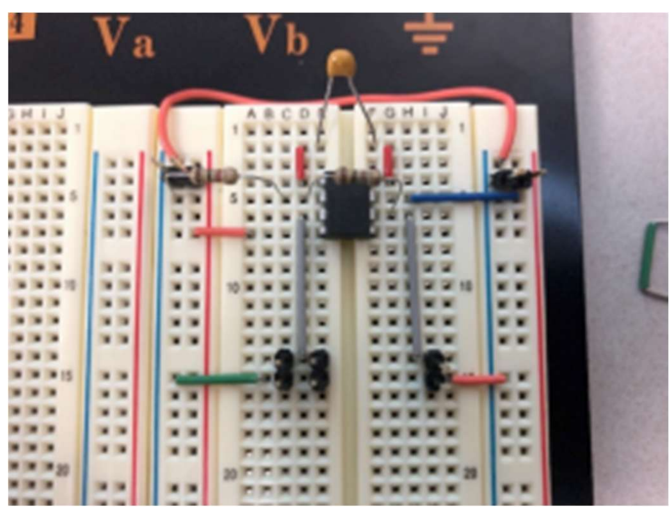


Figure 4 – Wiring Example

Figures 3 and 4 display the Pin-Out used for both op amps in this lab and wiring neatly is recommended to assist in debugging. Pins 4 and 7 (Fig.3) are the power supply pins. Connect pin 4 to $V_- = -5V$ (white wire) and pin 6 to $V_+ = +5V$ (red wire), Figure 1 may be helpful to reference. Pin 5 should be left floating (no connection). Attached is a ~56 pF capacitor between pins 1 and 8 of the op amp (if you do not have the exact capacitor value use a slightly larger one). This capacitor is known as **compensation capacitor** which is not needed for the LM741 op amp but necessary to stabilize the CA3130 op amp. You may find it useful to test the CA3130 with and without the compensations capacitor to observe the effects of it. *Note : When breadboarding, don't ground the output of your op amp unless specified.

I. Obtain the equations for the gain of both the inverting and non-inverting amplifiers. Reference figures 1 and 2 for schematics of op amp configuration for each. **(2 Points)**

$$Non-Inverting: 1+\frac{V_{out}}{V_{in}} = 1+\frac{R_2}{R_1} \mid Inverting: -\frac{V_{out}}{V_{in}} = -\frac{R_2}{R_1}$$

II. Let's take a look at the non-inverting amplifier circuit. We will assume $R_1 = 1\ k\Omega$ and want to achieve a gain of 11. Analytically find the correct value of R_2 to meet these specifications and build the circuit on your breadboard using the LM741 op amp. Test your circuit by applying a 1 kHz, 200 mV amplitude sine wave as your input. Include images of your work, breadboard, and oscilloscope. **(3 Points)**

$$Non-Inverting: V_{in}\left(1+\frac{R_2}{R_1}\right) \rightarrow \frac{R_2}{1k} = 10k \rightarrow R_2 = 10k$$

III. For the inverting amplifier circuit, we can keep $R_1 = 1\ k\Omega$, and but are instead looking to achieve a gain of -10. Repeat the same steps from part II. **(3 Points)**

$$Inverting: V_{in}\left(-\frac{R_2}{R_1}\right) \rightarrow -\frac{R_2}{R_1} = -10 \rightarrow -\frac{R_2}{1k} = -10 \rightarrow$$

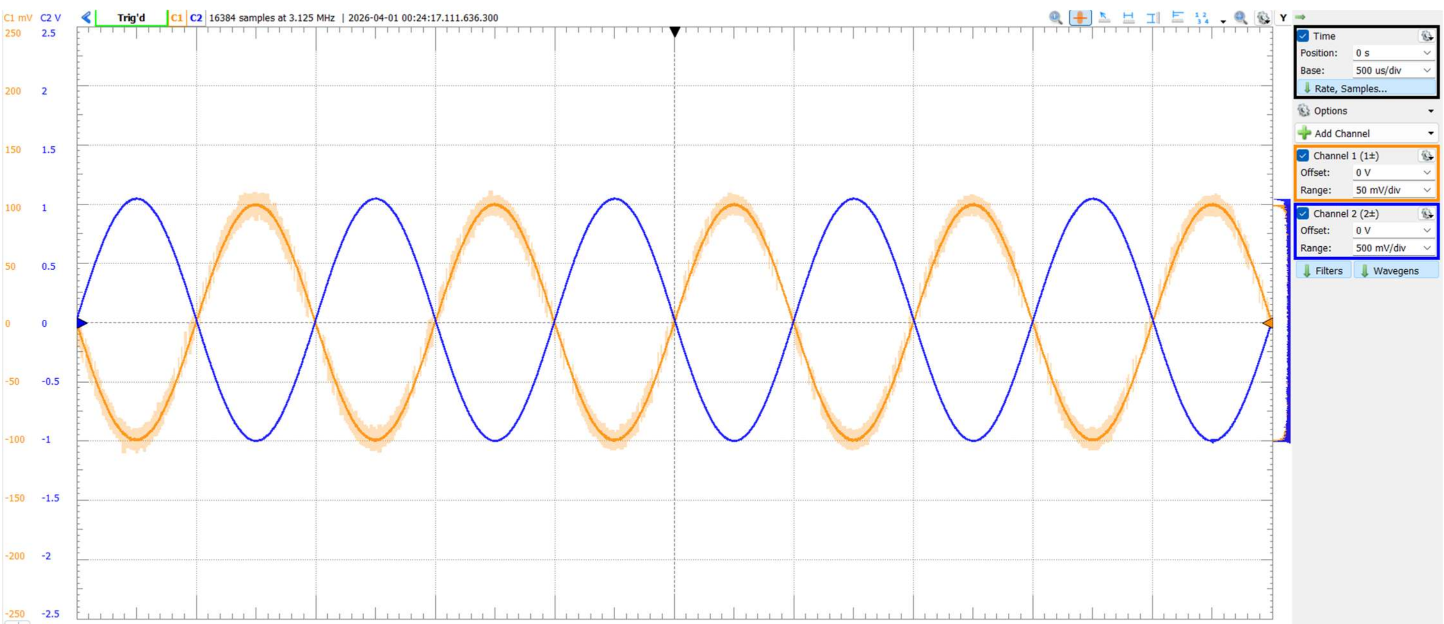
IV. Calculate by hand the input resistance of both the inverting and non-inverting amplifiers assuming the op amp is ideal. **(2 Points)**

For inverting, the op amp sees an infinite resistance and for a non-inverting it sees the input resistor of 1k.

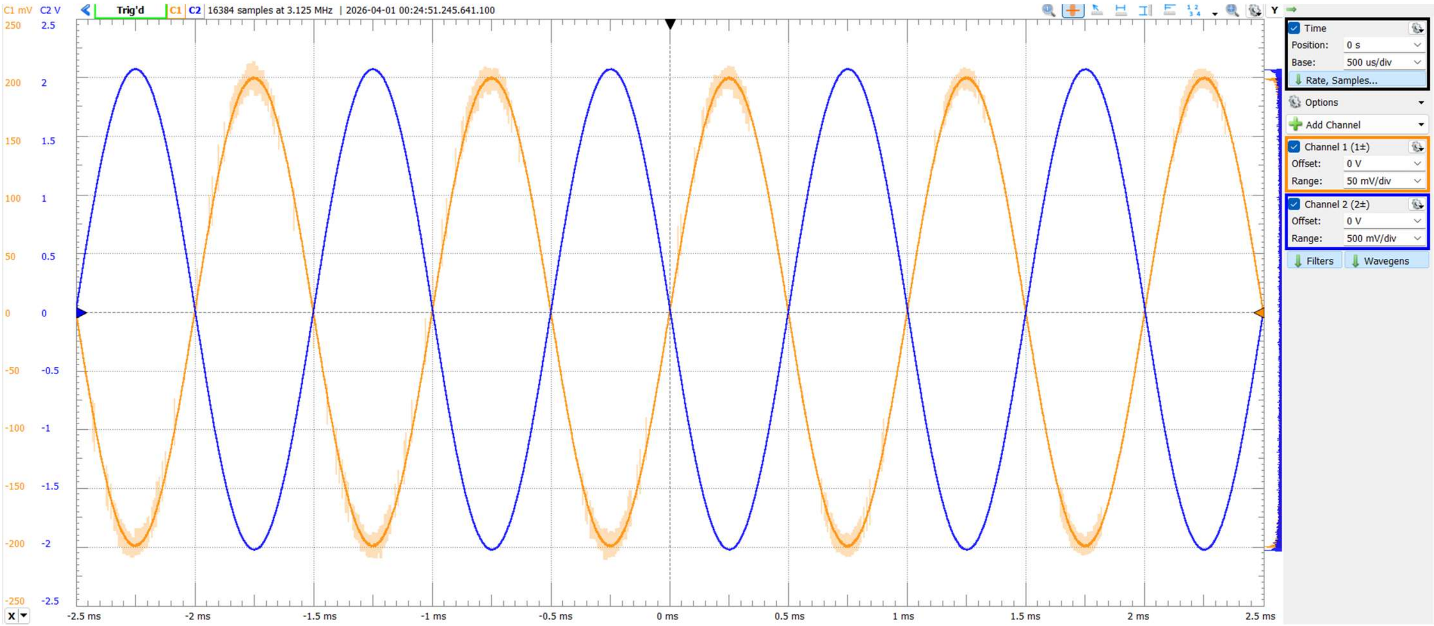
Part 2: Op Amp Output Voltage Swing (20 Points)

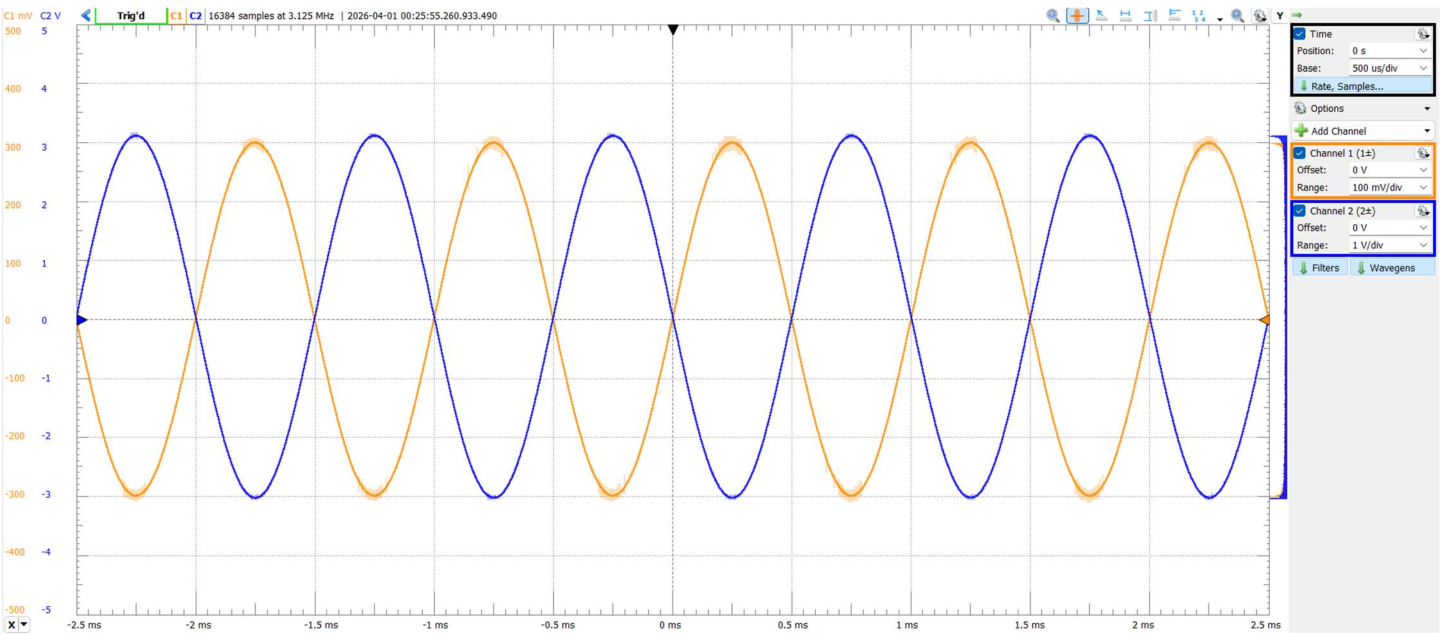
Ideal operational amplifiers are often assumed to produce output voltages that extend fully to the supply rails. Real devices, however, are limited by internal circuitry and load conditions, resulting in a restricted **output voltage swing**. When the output approaches these limits, the amplifier enters saturation and can no longer accurately reproduce the desired signal. The maximum and minimum achievable output voltages are experimentally determined and compared to theoretical expectations, highlighting the impact of supply voltages and load resistance on op-amp performance.

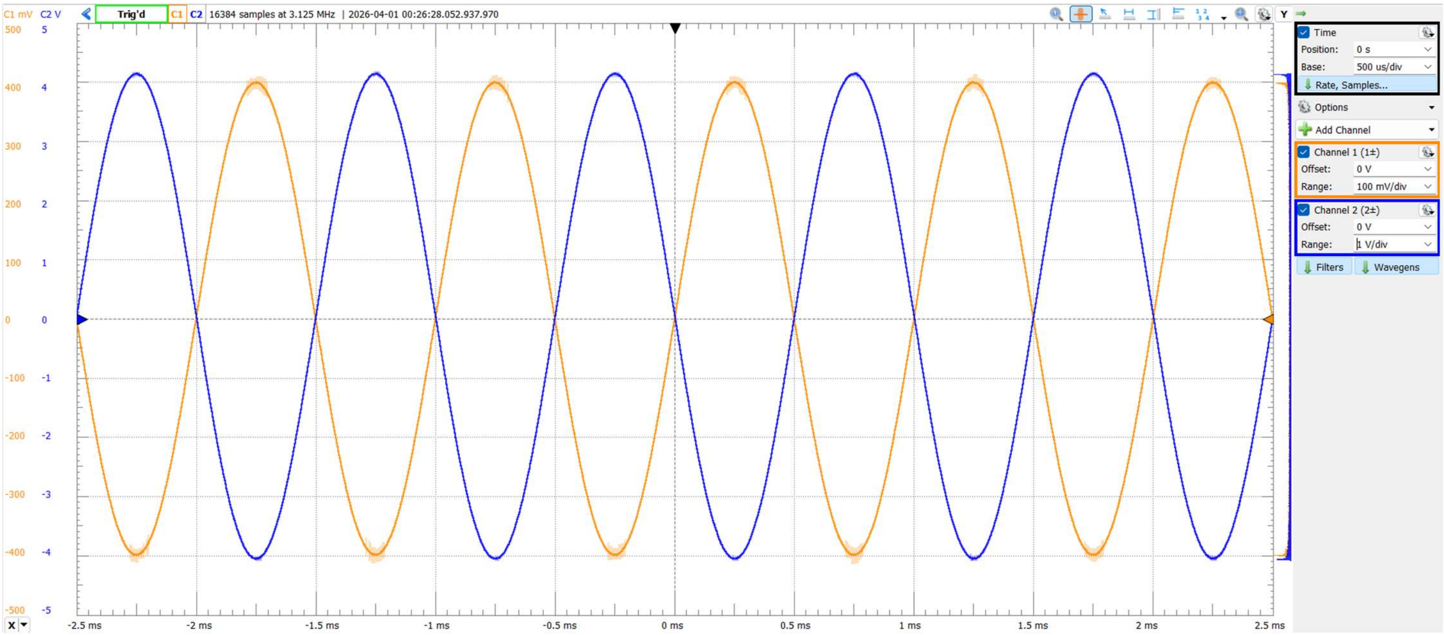
I. For this section build an inverting op amp using your LM741 with a $R_1 = 10\ k\Omega$ and $R_2 = 100\ k\Omega$. Demonstrate that the circuit works by applying a 100 mV amplitude 1kHz sine wave and observing the scope output. Have your channel 1 scope on the input pin and your channel 2 scope on the output pin. Screenshot your output. **(3 Points)**

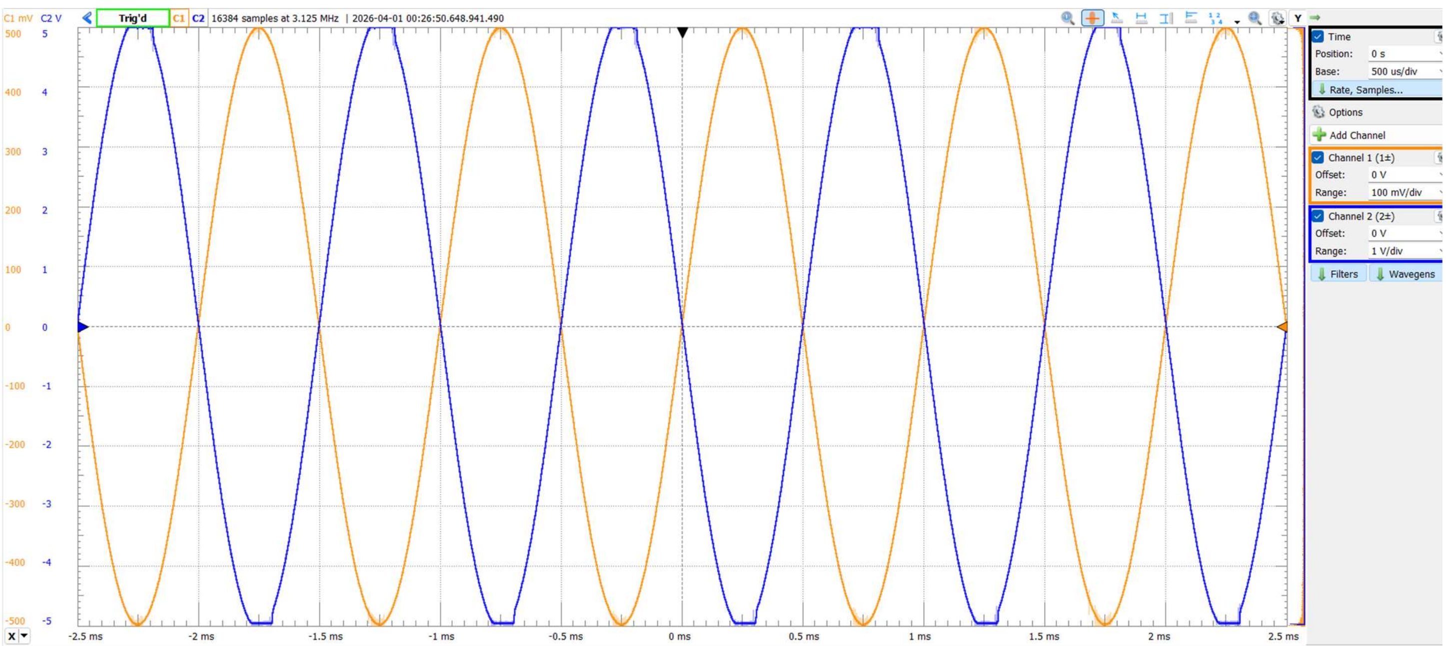


II. Repeat this step for an input amplitude of 200 mV, 300 mV, 400 mV, and 500 mV. Observe and explain what is happening with the output of the circuit. **(2 Points)**


Trig'd
16384 samples at 3.125 MHz | 2026-04-01 00:24:51.245.641.100
Time
Position: 0 s
Base: 500 us/div
Channel 1 (1±)
Offset: 0 V
Range: 50 mV/div
Channel 2 (2±)
Offset: 0 V
Range: 500 mV/div


Trig'd
16384 samples at 3.125 MHz | 2026-04-01 00:25:55.260.933.490
Time
Position: 0 s
Base: 500 us/div
Channel 1 (1±)
Offset: 0 V
Range: 100 mV/div
Channel 2 (2±)
Offset: 0 V
Range: 1 V/div

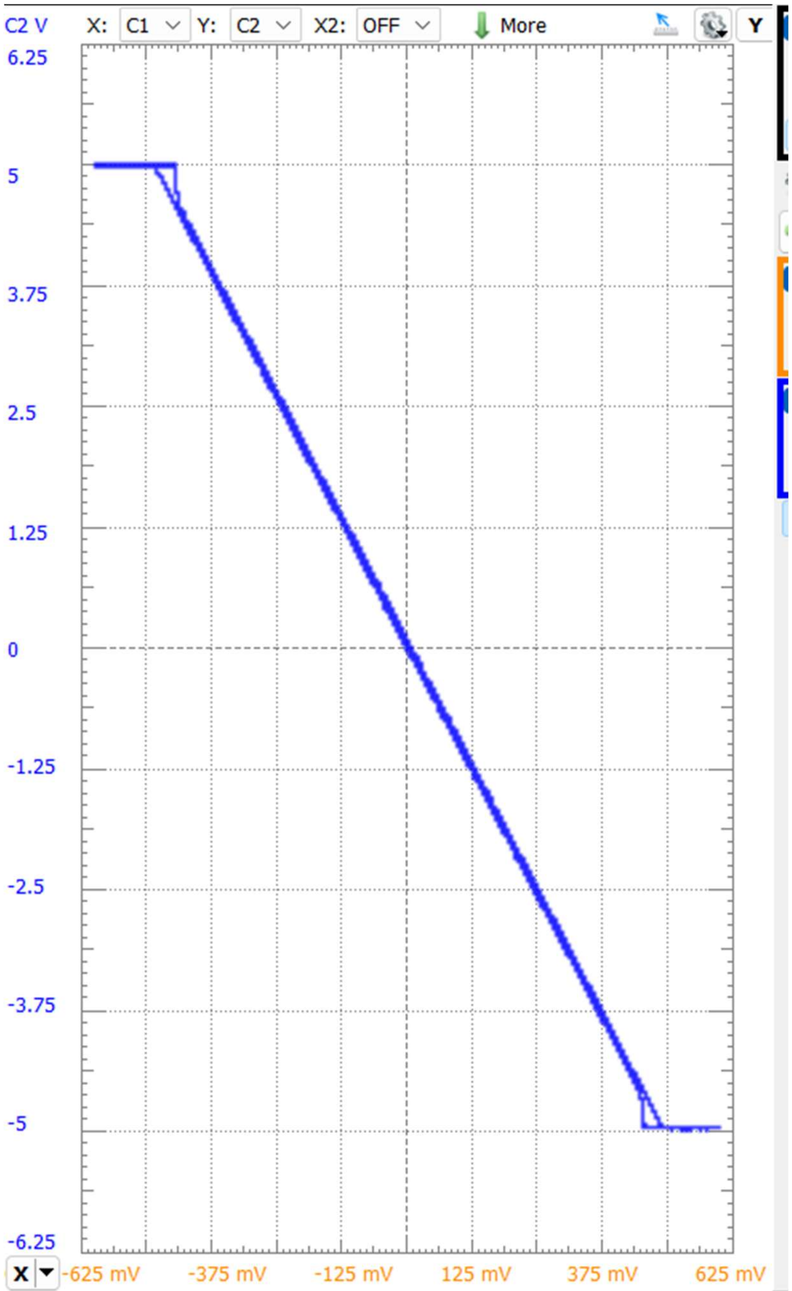




Maintains constant gain of 10 so long as frequency is constant.

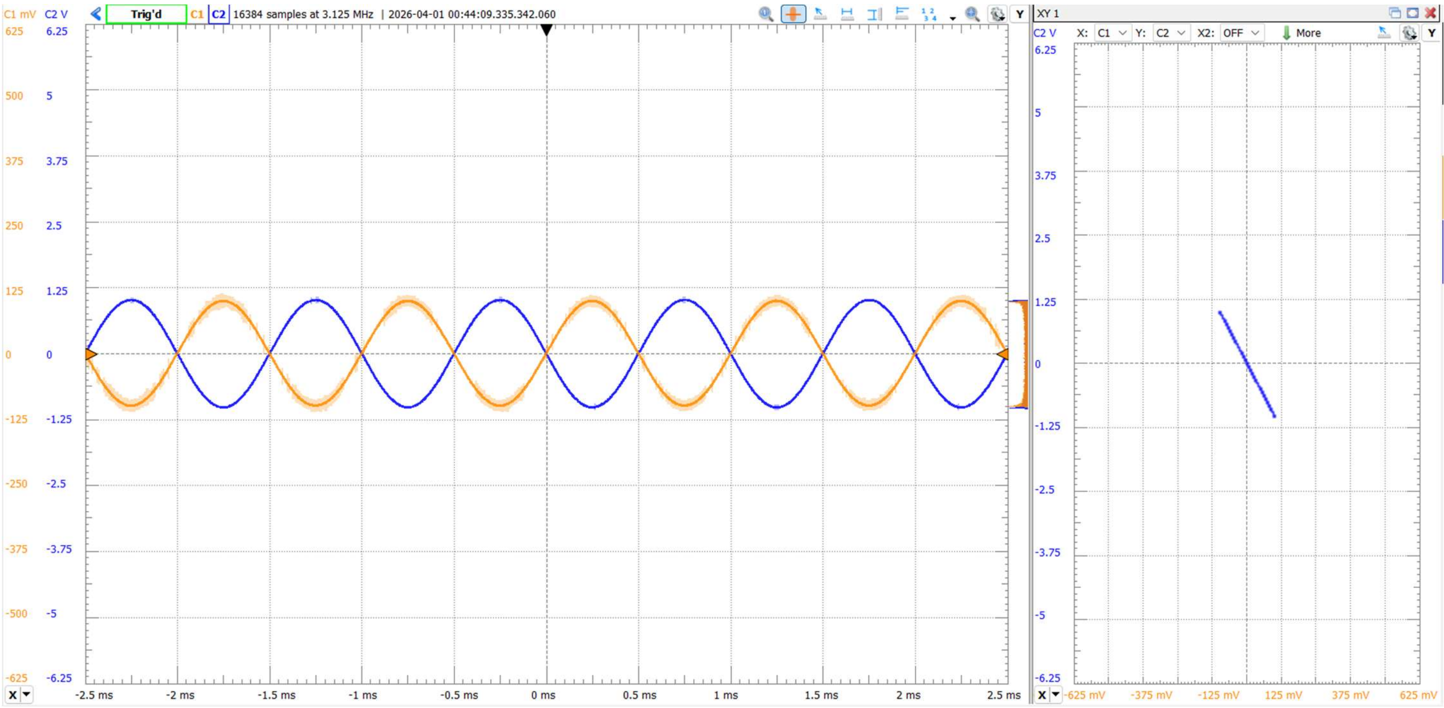
III. Use the XY Mode to plot the relationship between the input and output. First, increase the amplitude of the circuit to 600 mV. Then in scope, you will see a “+XY” button, click that to obtain an XY plot. Ensure that your Channel 1 (input) is on the x-axis and your Channel 2 (output) is on the y-axis. Screenshot your plot. **(4 Points)**

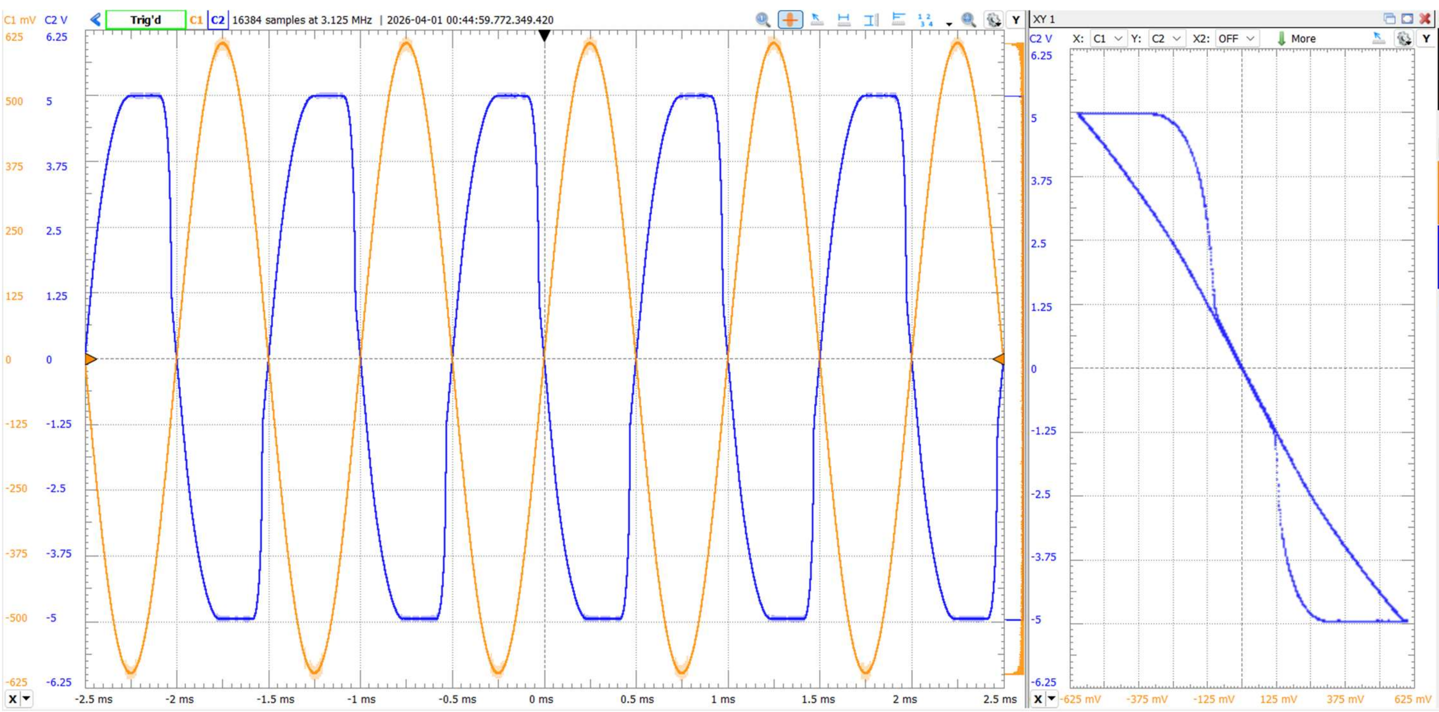
IV.



V. Repeat steps II and III with your CA3130. The pinout is the same as the LM741 so no re-wiring should be necessary. **(9 Points)**

Had to add a capacitor to give unity gain.



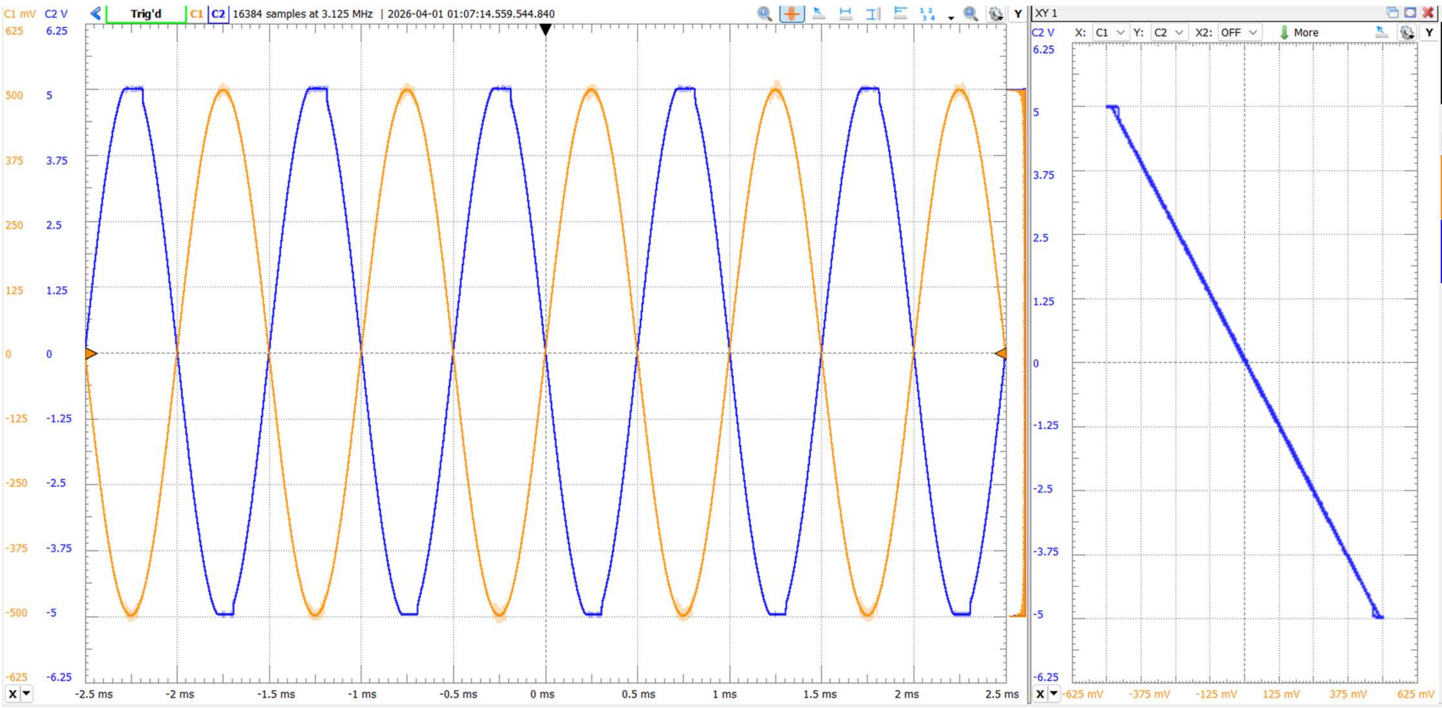


100mV vs 600mV in.

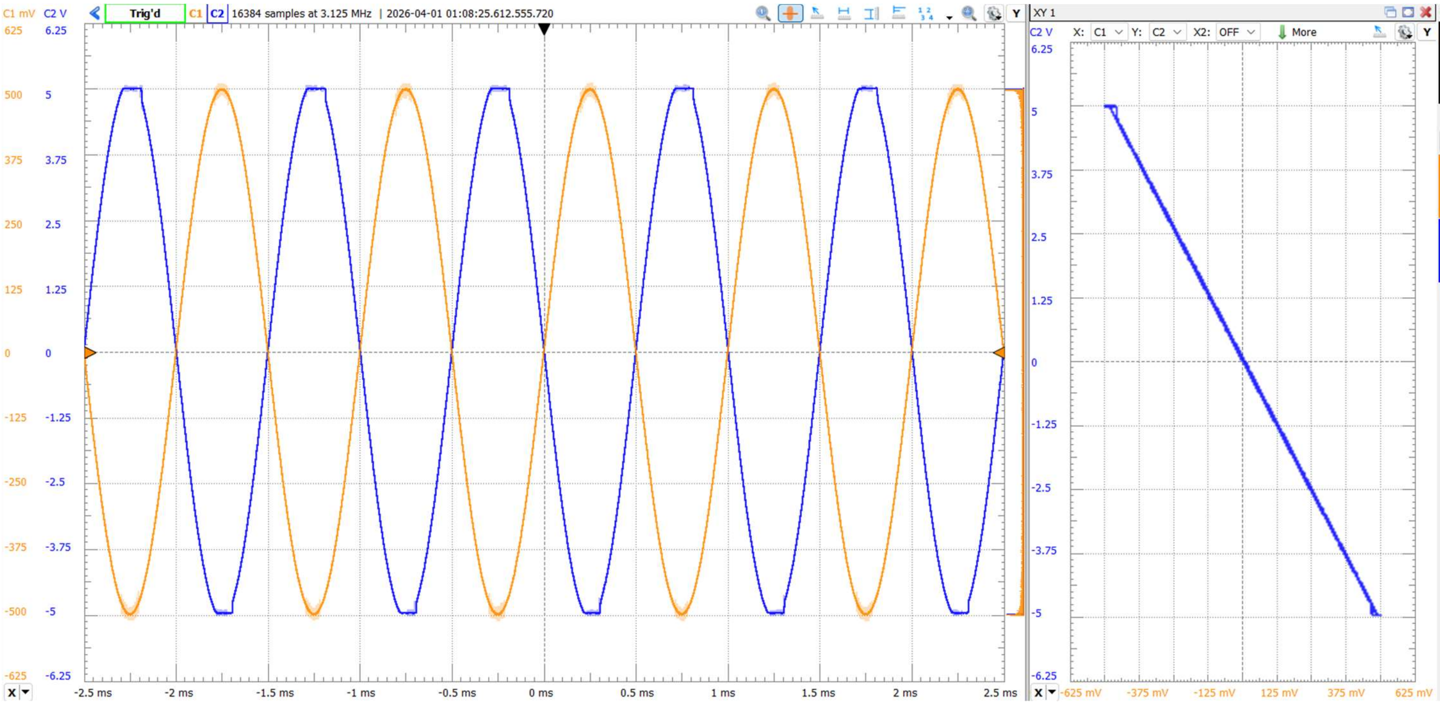
VI. Explain the difference between the voltage swing for the LM741 and the CA3130 and which one you would choose, provide reasoning and reference the data you just obtained. **(1 Point)**

I would use the CA3130 for high-frequency signals as it is made with a higher frequency tolerance, and for better rail-to-rail performance. I would use the LM741 for better voltage ranges.

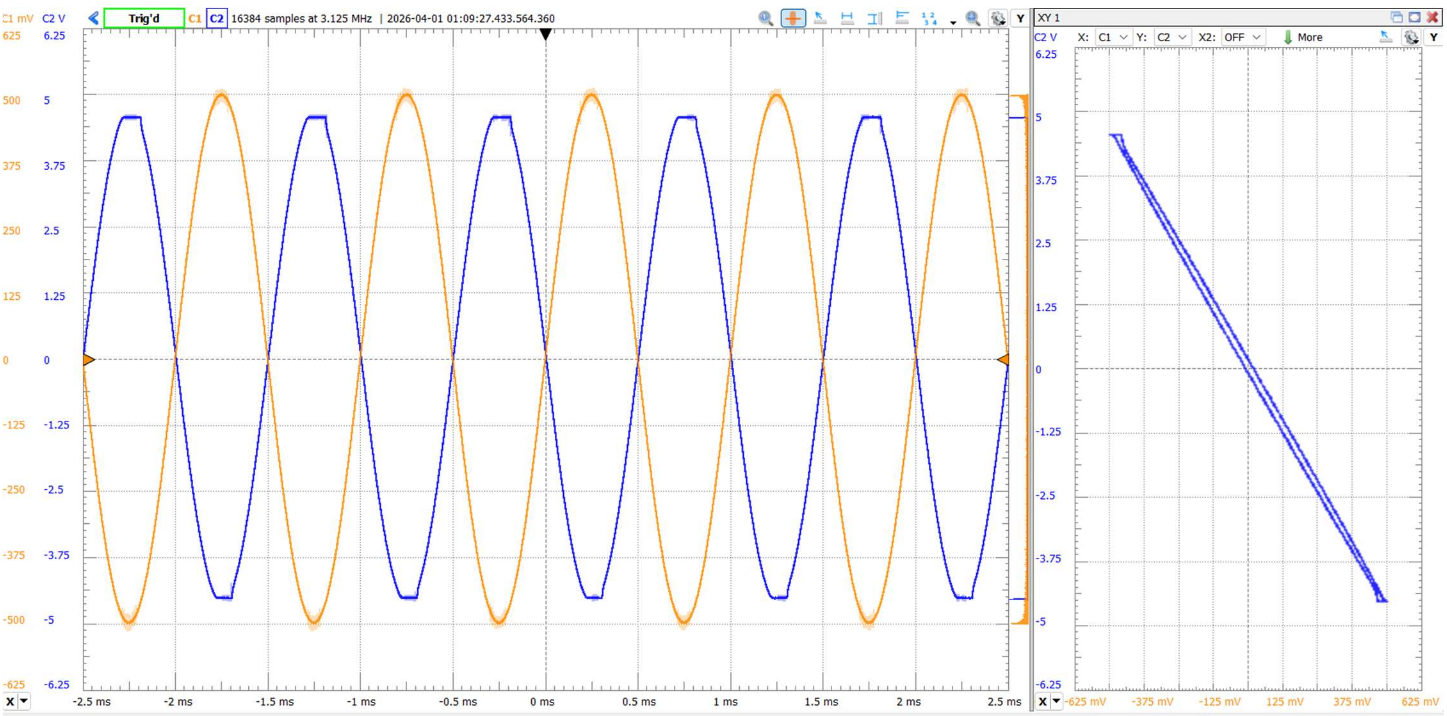
VII. Add a load resistor (R_L) to the output of the op amp (LM741). Observe any effects of adding the resistor to the circuit and explain what is happening. Use the following values for the load resistor: R_L = 100 Ω, 1 kΩ, 10 kΩ, and 100 kΩ resistor. **(1 Point)**



100


C1 mV C2 V
Trig'd
C1 C2 16384 samples at 3.125 MHz | 2026-04-01 01:08:25.612.555.720
625 6.25
500 5
375 3.75
250 2.5
125 1.25
0 0
-125 -1.25
-250 -2.5
-375 -3.75
-500 -5
-625 -6.25
-2.5 ms -2 ms -1.5 ms -1 ms -0.5 ms 0 ms 0.5 ms 1 ms 1.5 ms 2 ms 2.5 ms
XY 1
C2 V X: C1 Y: C2 X2: OFF More
-625 mV -375 mV -125 mV 125 mV 375 mV 625 mV

10k

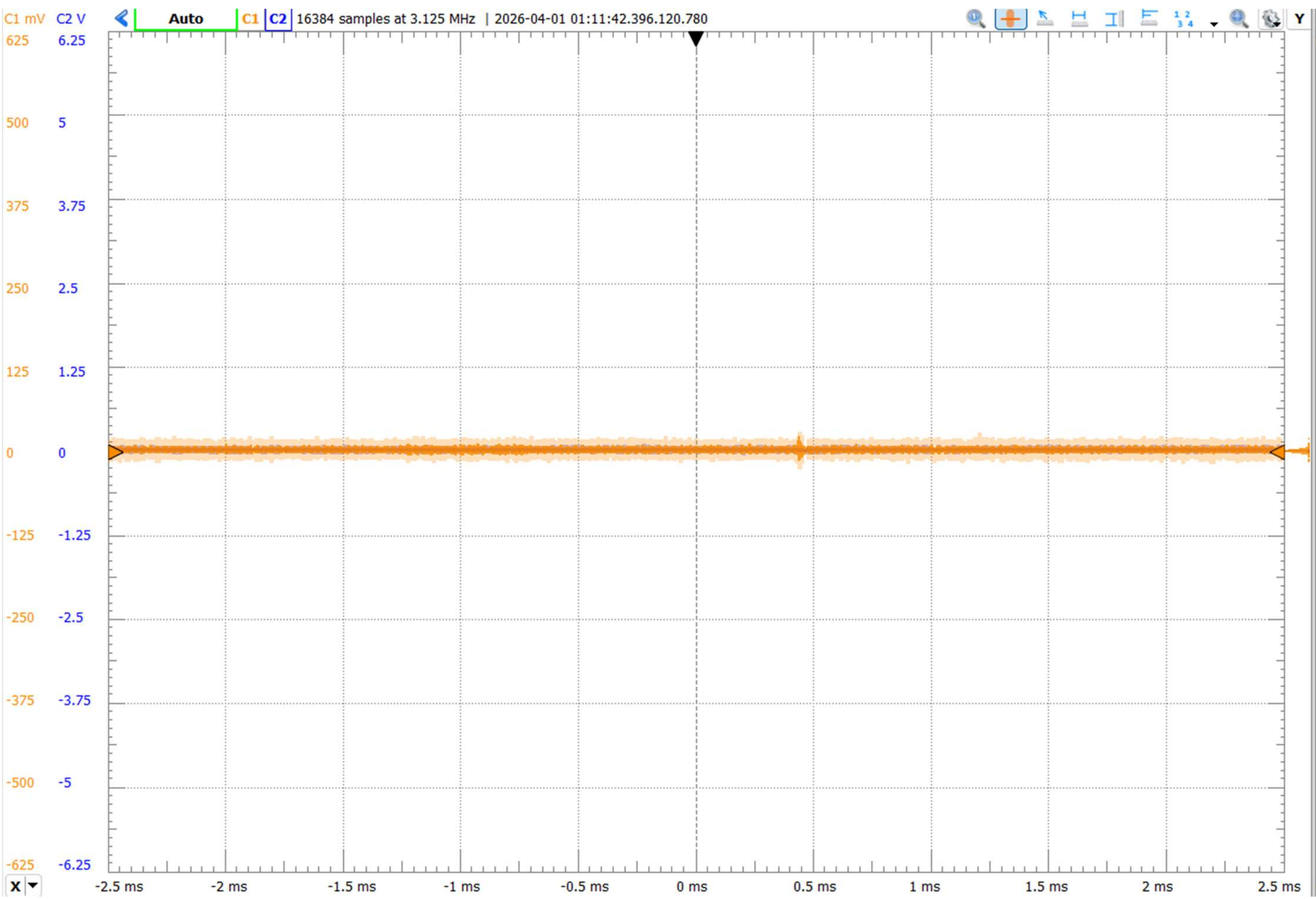

C1 mV C2 V
Trig'd
C1 C2 16384 samples at 3.125 MHz | 2026-04-01 01:09:27.433.564.360
625 6.25
500 5
375 3.75
250 2.5
125 1.25
0 0
-125 -1.25
-250 -2.5
-375 -3.75
-500 -5
-625 -6.25
-2.5 ms -2 ms -1.5 ms -1 ms -0.5 ms 0 ms 0.5 ms 1 ms 1.5 ms 2 ms 2.5 ms
XY 1
C2 V X: C1 Y: C2 X2: OFF More
-625 mV -375 mV -125 mV 125 mV 375 mV 625 mV

100k

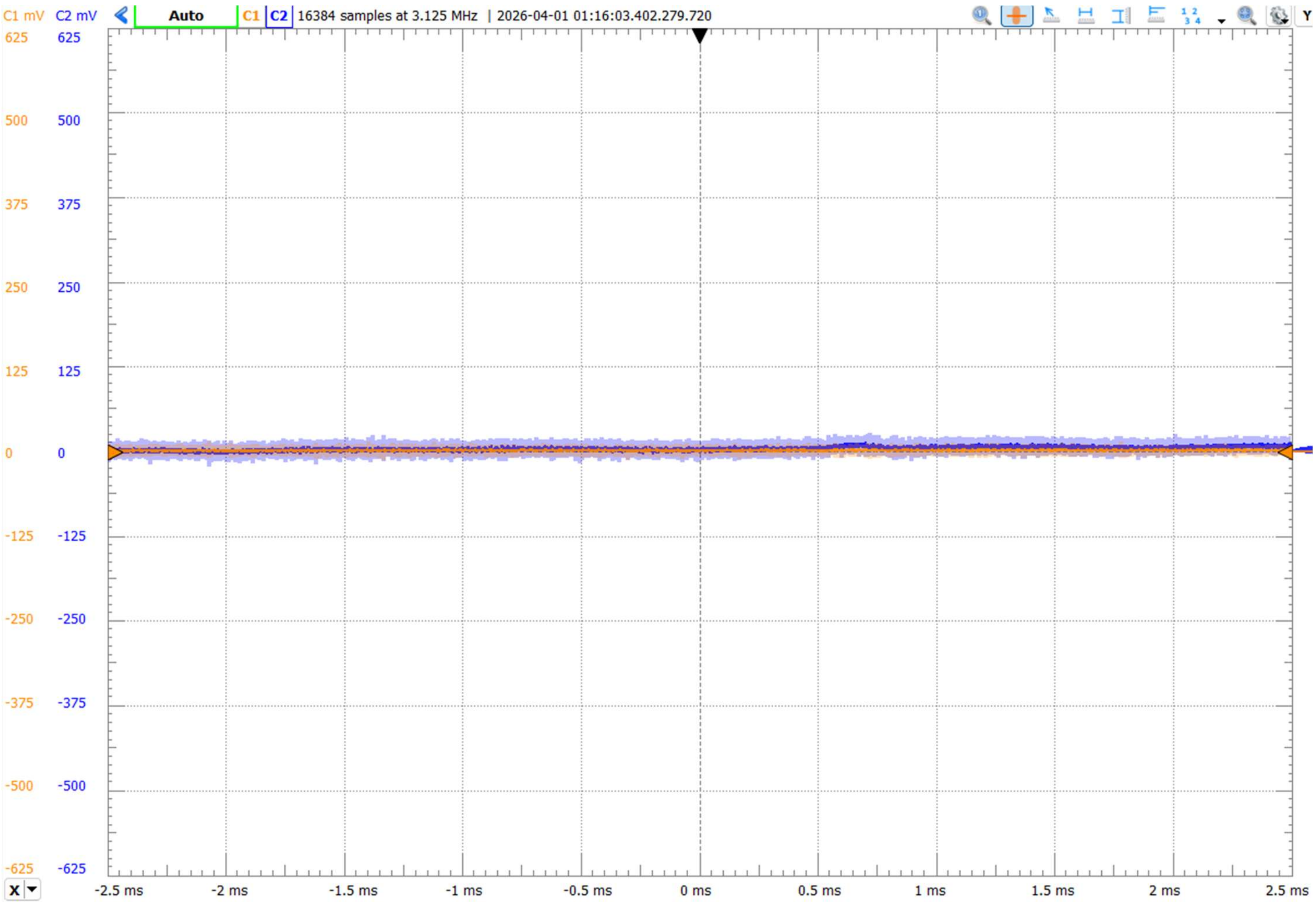
Part 3: Input Offset Voltages (20 Points)

This section will analyze how even when there is no input to a circuit, there will still be some output to the circuit due to imperfections of the op amp. In an ideal operational amplifier, having no input voltage would result in an output of zero. In practical op-amps, internal device mismatches cause a small DC voltage to appear at the input, known as the **input offset voltage**. Although typically only a few millivolts or less, this offset can be significantly amplified in closed-loop configurations, resulting in measurable output error. In section, the input offset voltage of the operational amplifier is determined experimentally, illustrating how small non-idealities at the input can affect overall circuit accuracy in precision analog applications.

I. Configure an inverting op amp circuit with your LM741 with R_1 = 10 kΩ and R_2 = 100 kΩ. Ensure that the input of the circuit is grounded. Measure the amplitude of the output using the scope measurements feature. Note that you may see a lot of "fuzziness" in the V_{out} waveform. This is due to noise, which is a limitation of all amplifiers. **(1 Point)**



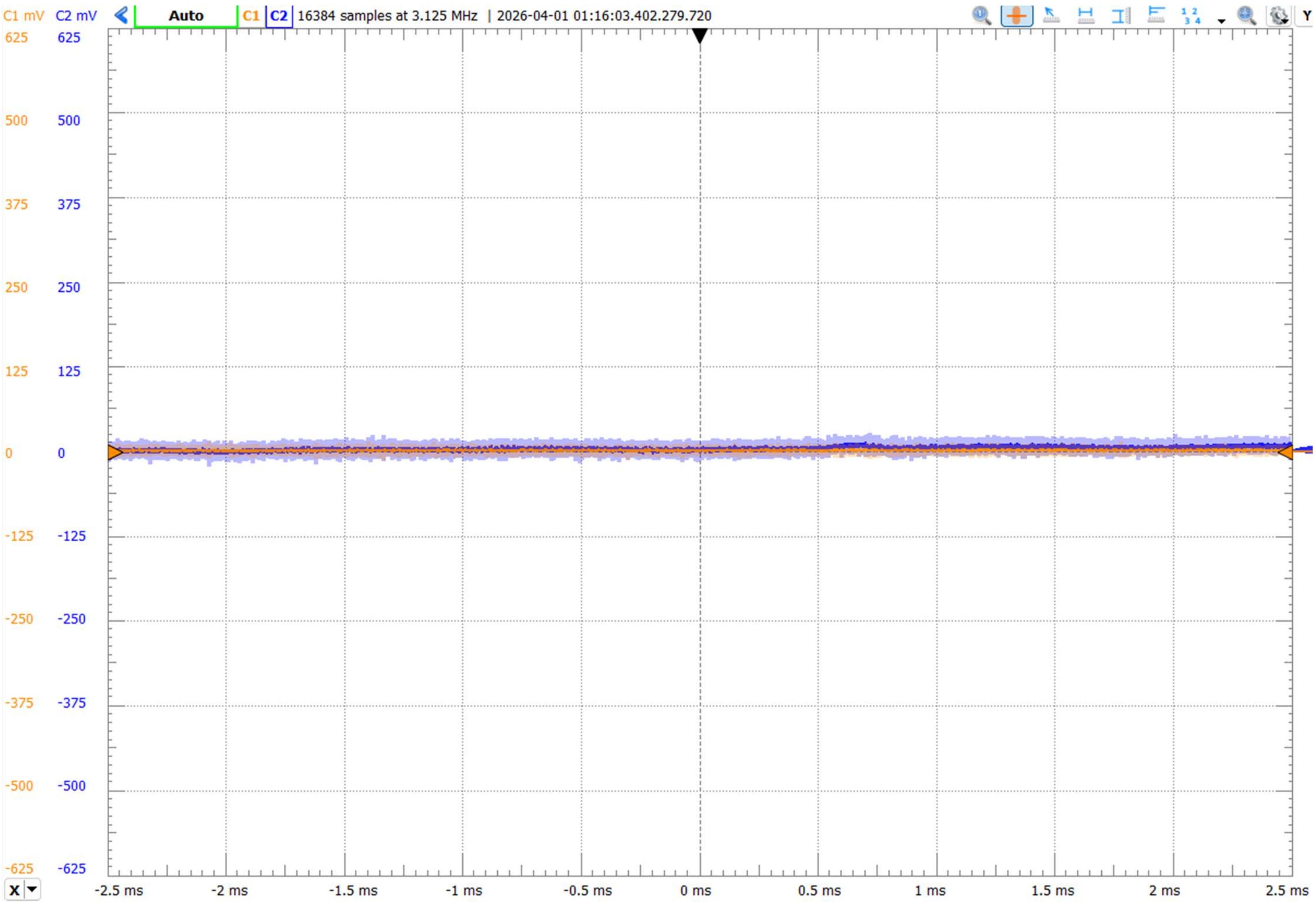
II. Now replace R_1 with a 1 kΩ resistor and measure the output with this type of gain. **(3 Points)**



III. You should see a small output voltage on the scope. For an ideal op-amp, this value would be zero. Measure this value (for R_1 = 1 kΩ) and calculate $\mathbf{V_{OS}}$ using the formula below. **(3 Points)**

$$V_{OS} = 0.00000858762$$

IV. Repeat Steps II and III with the CA3130 op Amp. **(6 Points)**



V. Again, based on the results of the last few steps, which op amp has better performance with respect to the following non-ideality we measured in this section. **(2 Points)**

$$V_{OS} = 0.00001014851$$

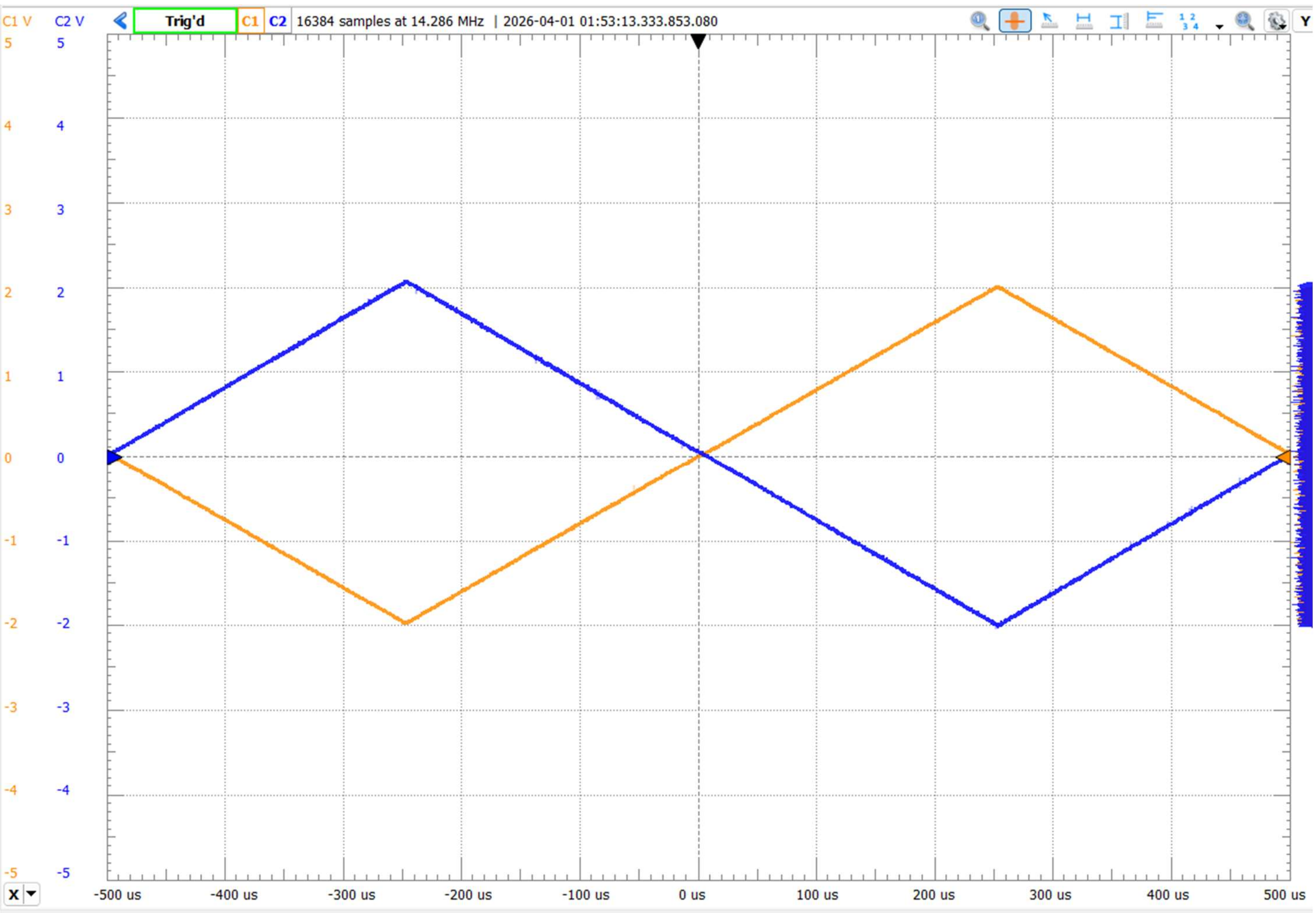
VI. Now change back to the LM741 and configure your input to a 0.5V amplitude 100 Hz triangular wave. To test the limits of how much current the op amp can provide, load V_{out} by placing a pair of 100 Ω resistors in parallel between V_{out} and ground. Measure the output voltage and calculate the current flowing through the equivalent 50 Ω resistor using Ohm's Law. **(5 Points)**

1.0100 V across 50 ohms -> 0.20002A

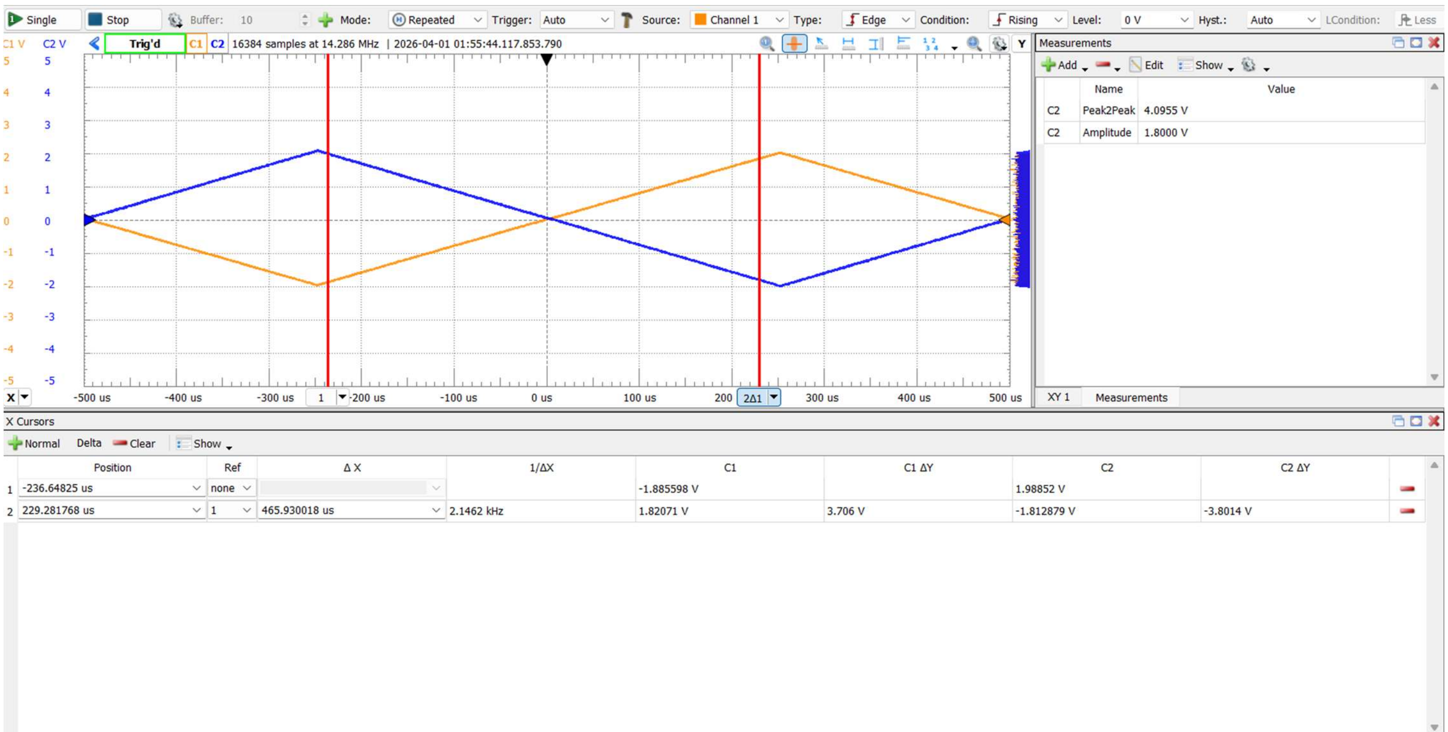
Part 4: Slew Rate (10 Points)

An ideal operational amplifier is assumed to respond instantaneously to changes in its input signal. In reality, the rate at which the output voltage can change is finite and is characterized by the **slew rate**. When driven by large or high-frequency signals, an op-amp may become slew-rate limited, resulting in signal distortion and reduced bandwidth. This section investigates the slew rate by observing the op-amp's response to rapidly changing input signals, demonstrating how this non-ideality constrains the amplifier's performance in time-varying applications.

I. Build an inverting amplifier with a gain of 1 using the LM741. Configure the input to be a square wave with a frequency of 1 kHz and an amplitude of 2 V. Observe the input and output waveforms on the scope. Decrease the time base so that most of the scope is focused on the transition from high to low on the output. Include a screenshot of your output in your report. Also compare the rise/fall times of the input and output. **(2 Point)**



II. Measure the slew rate (the slope of the output waveform when the output changes from high to low), you can utilize certain features of waveforms to easily measure this. Compare this to the expected slew rate of this part (check datasheet). **(1 Points)**

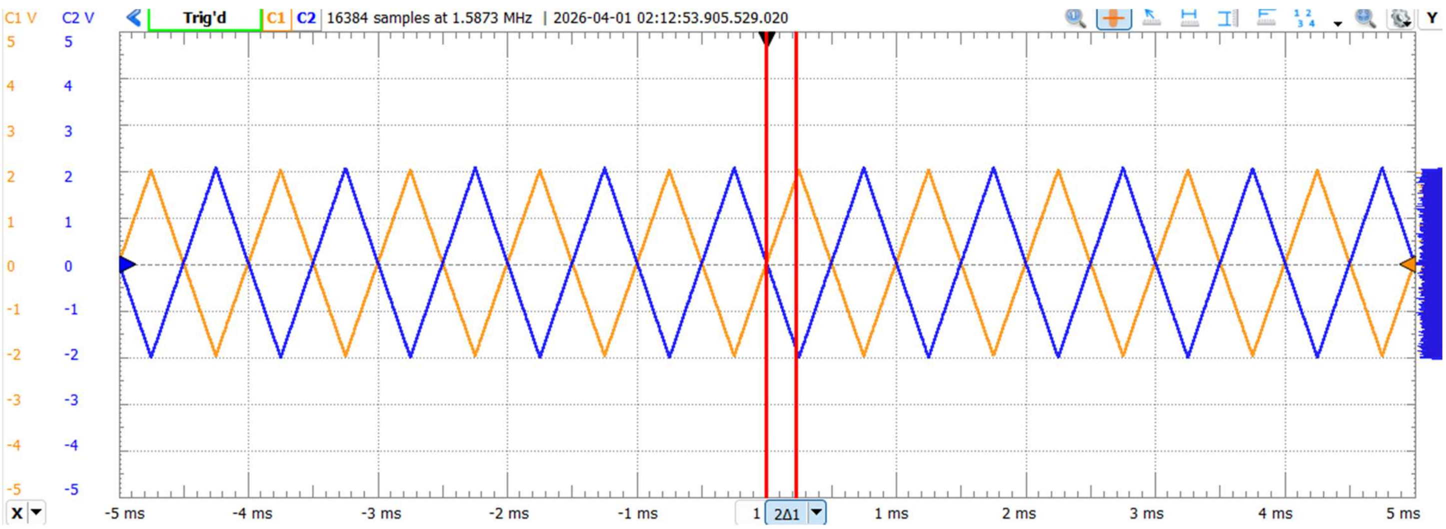


0.00811066008 V/us, this is much lower than the limitations due to it being well below the maximum tolerated by the lm741.

III. Repeat this experiment comparing the slew rate of the op amp based on the frequency by obtaining the slew rate of the op amp when the input frequency is 100 Hz, 10 kHz, 100 kHz, and 1 MHz. Reflect on the effects of varying the input frequency across this range with respect to the output. **(4 Points)**

10kHz: 0.03894689515V/us
100kHz: 0.419918369 V/us
1MHz: 0.509228811 V/us

IV. Replace the LM741 with the CA3130EZ. Repeat steps I and II for this part. **(3 Points)**



0.00813152792 V/us

10kHz: 0.03774620051 V/us

100kHz: 0.45434225586 V/us

1MHz: 3.37011494253 V/us

Much higher slew rate max.

Part 5: Active Filter (20 Points)

Although an active filter isn't a non-ideality of an op amp, it will have some parallelism in the In-Lab portion. It is also an important concept to understand for the course so it will be practiced in this section. Feel free to use either (or both) op amps in this section. You are to only design using an inverting op amp configuration but can choose the resistor/capacitor values. Feel free to ask any questions if necessary. Note that a 3% error is allowed for the specified requirements in the following section with respect to cut off frequency and gain.

In terms of designing your circuits, these schematics and equations should be all you need for reference. *Note that the equations for cutoff frequency and gain are the same for both the high pass and low pass circuits.

Cutoff Frequnecy: $f_c = \frac{1}{2\pi R} Hz$

Gain: $A_v = 1 + \frac{R_2}{R_1}$

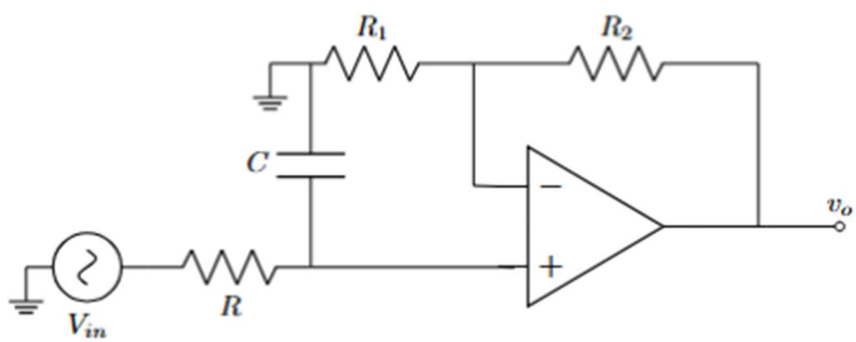


Figure 5 – Active Low Pass Filter Circuit

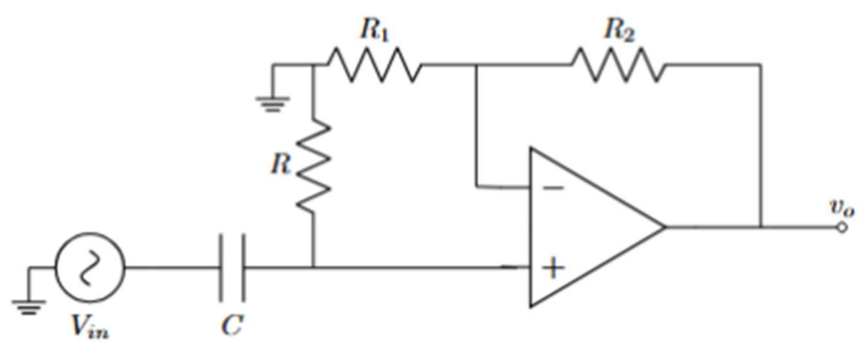
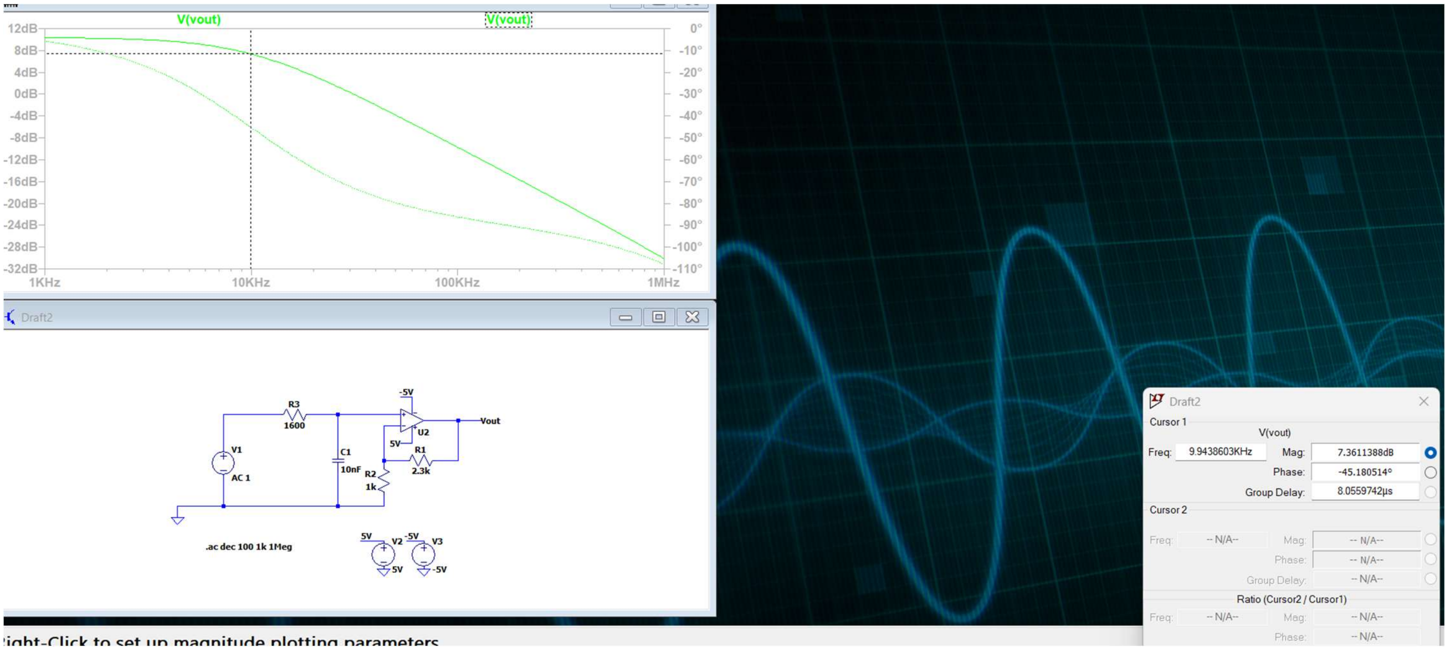


Figure 6 – Active Low Pass Filter Circuit

I. Design an active low-pass filter with a cut-off frequency of 10 kHz and a pass band gain of 3.3 V/V. **(5 Points)**

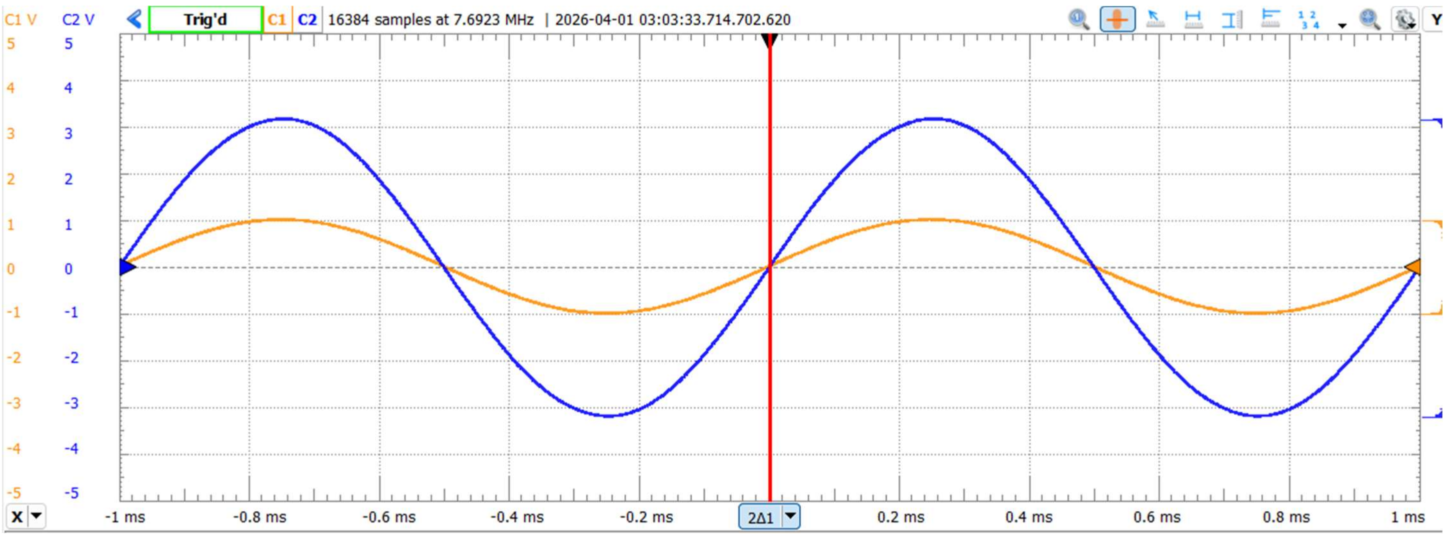


II. Create this circuit in LT Spice (Use universal op amp II or import an LM741 part) and test it out using both timing analysis and frequency analysis. **(2 Points)**

For frequency analysis in LTspice:

- Right click your input voltage source, select "Advanced" and set its AC amplitude to 1V, and leave other settings as default.
- Go to Simulate > Edit Simulation Cmd.
- Select the AC Analysis tab.
- Set the type of sweep to Decade, choose the number of points per octave/decade (e.g., 151), and specify the frequency range that would test all relevant frequencies (e.g., 10Hz to 1MHz).
- Run the Simulation.

III. Build this circuit on the breadboard and test the output of the system. **(2 Points)**



Shows gain at 1kHz is about 3.3, lowers as frequency increases.

IV. Run the circuit through the network analyzer to obtain the frequency response. **(1 Point)**

For the network analyzer of the DAD:

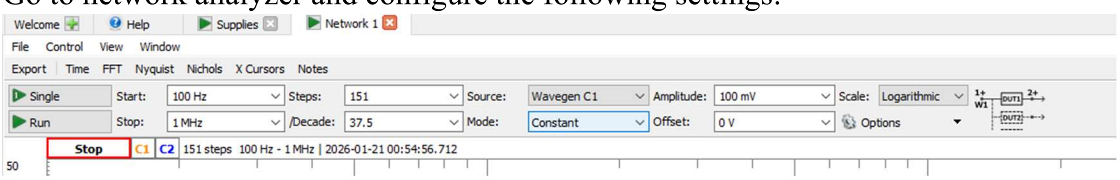
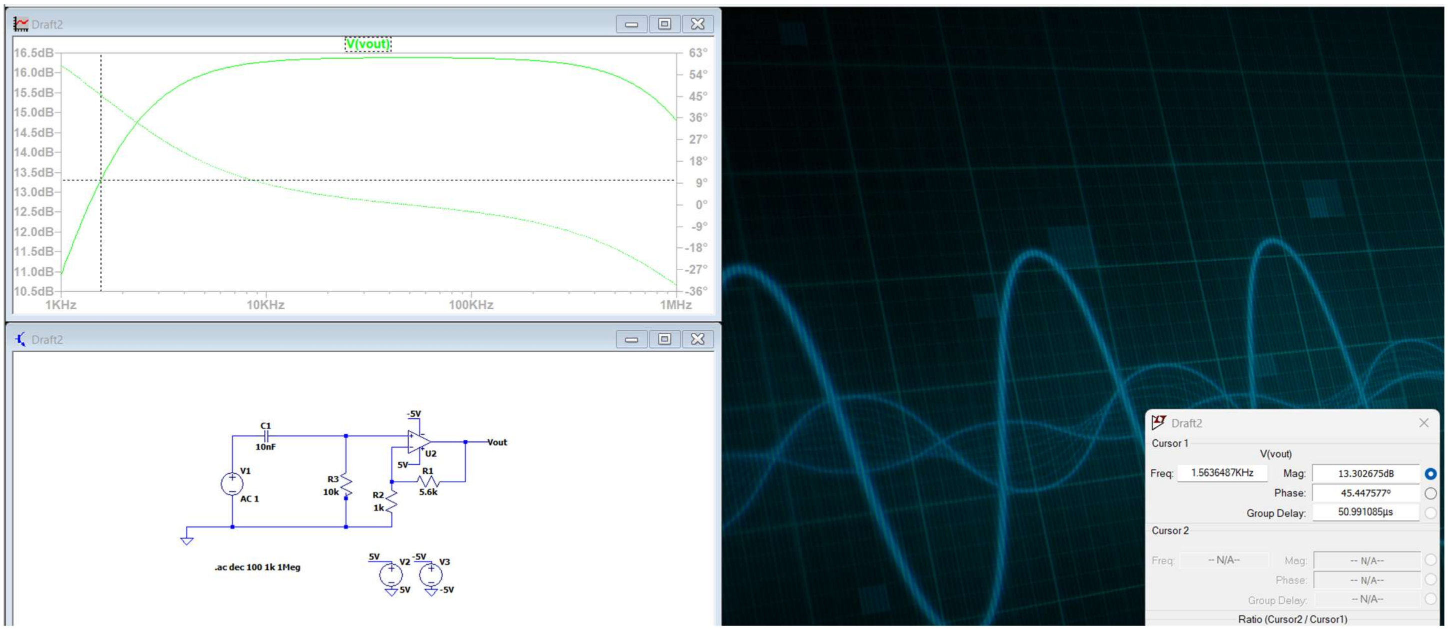
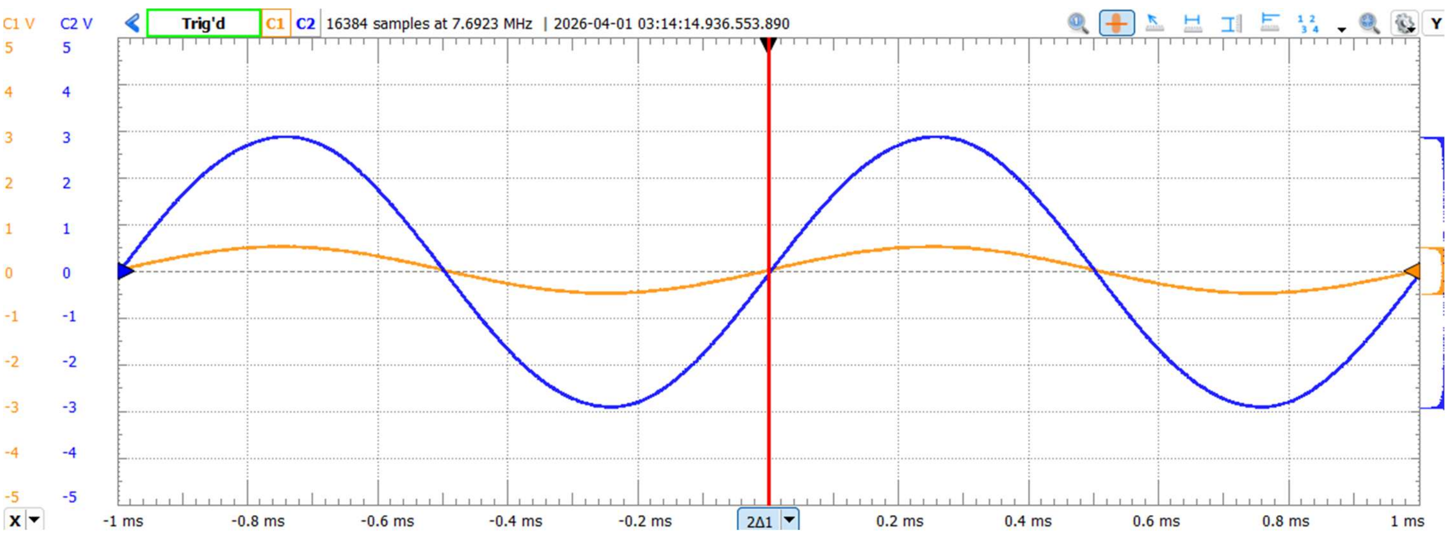
- Ensure all power/supplies are being supplied to op amp (along with proper grounding).
- Ensure channel 1 is on the input pin (with Wavegen 1) and channel 2 is on the output pin.
- Go to network analyzer and configure the following settings:
- 

Figure 7 – Network Analyzer Settings

- Run the network analyzer to obtain your output.

V. Repeat steps I-IV for an active high pass filter with a cut-off frequency of 1.5 kHz and a pass band gain of 6.7 V/V. **(10 Points)**


Draft2
V(vout)
16.5dB
16.0dB
15.5dB
15.0dB
14.5dB
14.0dB
13.5dB
13.0dB
12.5dB
12.0dB
11.5dB
11.0dB
10.5dB
63°
54°
45°
36°
27°
18°
9°
0°
-9°
-18°
-27°
-36°
1KHz
10KHz
100KHz
1MHz
C1
10nF
V1
AC 1
R3
10k
R2
1k
R1
5.6k
U2
-5V
5V
Vout
.ac dec 100 1k 1Meg
V2
V3
Cursor 1
V(vout)
Freq: 1.5636487KHz
Mag: 13.302675dB
Phase: 45.447577°
Group Delay: 50.991085µs
Cursor 2
Freq: --N/A--
Mag: --N/A--
Phase: --N/A--
Group Delay: --N/A--
Ratio (Cursor2 / Cursor1)


C1 V
C2 V
Trig'd
C1
C2
16384 samples at 7.6923 MHz | 2026-04-01 03:14:14.936.553.890
5
4
3
2
1
0
-1
-2
-3
-4
-5
-1 ms
-0.8 ms
-0.6 ms
-0.4 ms
-0.2 ms
2Δ1
0.2 ms
0.4 ms
0.6 ms
0.8 ms
1 ms

HPF at 1kHz